

TASK-5

Writing Join Queries, Equivalent AND/OR Recursive Queries

Topic: Implementation of different types of joins and recursive queries

- A SQL join combines records from two tables
- A join locates related column values in the two tables
- A query can contain zero, one, or multiple join operations.
- Inner join is the same as join, the keyword Inner is optional

Objective :-

To implement different types of joins and recursive queries

Theory:-

The SQL join clause is used to combine records from two or more tables in a database. A join is a means for combining fields from two tables by using values common to each. The join is actually performed by the 'where' clause which combines specified rows of tables.

Syntax:-

select column1, column2, column3 from table-name1, table-name2
where table-name1.columnname = table-name2.columnname;

Types of Joins:

1. Simple Join
2. Self Join
3. Outer Join

Simple Join:

It is the most common type of join. It retrieves the rows from 2 tables having a common column and is further classified into

Equi-join:

A join, which is based on equality, is called equi-join.

Example:

select * from item, cust where item.id = cust.id;

In the above statement, item.id = cust.id performs the join statement. It retrieves rows from both the tables provided they both have the same id as specified by the where clause. Since the where clause uses the comparison operator (=) to perform a join, it is said to be equi-join. It combines the matched rows of tables. It can be used as follows:

- To insert records in the target table.
- To create tables and insert records in this table.
- To update records in the target table.
- To create views.

Non Equi-join:

It specifies the relationship between columns belonging to different tables by making use of relational operators other than '=', '<', '>'.

Example:-

select * from item, cust where item.id < cust.id;

Table Aliases.

Table aliases are used to make multiple table queries

shorted and more readable. We give an alias name to the table in the 'from' clause and use it instead of the name throughout the query.

Self join:

Joining of a table to itself is known as self-join. It joins one row in a table to another. It can compare each row of the table to itself and also with other rows of the same table.

Example:

select * from emp1, emp2 where v1.salary > (select avg(salary) from emp2 where v1.deptno = v2.deptno);

Outer Join:-

It extends the result of a simple join. An outer join returns all the rows returned by simple join as well as those rows from one table that do not match any row from the table. The symbol (+) represents outerjoin. Different types of SQL

Joins.

Here are the different types of the joins in SQL:

(INNER) JOIN: Returns records that have matching values in both tables

select column_name(s) from table1 INNER JOIN table2 on table1.
column_name = table2. column_name;

LEFT (OUTER) JOIN: Return all records from the left table and the matched records from the right table.

select column_name(s) from table1 left join table2 on table1.
column_name = table2. column_name;

RIGHT (OUTER) JOIN: Return all records from the right table and the matched records from the left table.

select column_name(s) from table1 right join table2 on table1.
column_name = table2. column_name;

Full Outer Join: Return all records when there is a match in either left or right table. Select column_name(s) from table1.

Full outer join table2 on table1. column_name = table2. column_name;
consider the following two tables. member + borrowed
Inner join every

select borrowed. membno, member. NAME from member
INNER JOIN borrowed
on member. membno = borrowed. membno;

Left Join Query

select member. NAME, borrowed. membno FROM member
left join borrowed on borrowed. membno = member. membno;

SQL Right Join keyword
select member. NAME, borrowed. membno FROM member
right join borrowed on borrowed. membno = member. membno;

SQL Full Outer Join keyword
select member. NAME, borrowed. membno FROM member
full outer join borrowed on borrowed. membno = member. membno;

TIP: Full outer join and full join are the same select

members, MEMBE1 borrowed. membno FROM member.
Full join borrowed on borrowed. membno = member. membno;

Syntax:-

with RECURSIVE x2 (result1 AS

SELECT 1

UNION ALL

SELECT result1 FROM x2;

SELECT * FROM x2 LIMIT 10;

VEL TECH	
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
SIGN WITH DATE	

21/08/2025

Result:- Thus, the writing join queries, Equivalent, and for recursive queries if done successfully

SQL> create table borrowed(mem_no number(2),book_id number(5));

Table created.

SQL> desc borrowed

Name	Null?	Type
MEM_NO		NUMBER(2)
BOOK_ID		NUMBER(5)

SQL> insert into borrowed values(2,'101');

1 row created.

SQL> insert into borrowed values(3,'102');

1 row created.

SQL> insert into borrowed values(5,'103');

1 row created.

SQL> select * from borrowed;

MEM_NO	BOOK_ID
2	101
3	102
5	103

SQL> select borrowed.mem_no,members.name from members inner join borrowed on
members.mem_no=borrowed.mem_no;

MEM_NO	NAME
2	Bob
3	charlie

SQL> select members.name,borrowed.mem_no from members left join borrowed on
borrowed.mem_no=members.mem_no;

NAME	MEM_NO
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SQL> connect

Enter user-name: system

Enter password:

Connected.

SQL> create table members(mem_no number(2),name varchar(15));

Table created.

SQL> desc members

Name	Null?	Type
MEM_NO		NUMBER(2)
NAME		VARCHAR2(15)

SQL> insert into members values(1,'alice');

1 row created.

SQL> insert into members values(2,'Bob');

1 row created.

SQL> insert into members values(3,'charlie');

1 row created.

SQL> insert into members values(4,'David');

1 row created.

NAME	MEM_NO
Bob	2
charlie	3
David	
alice	

SQL> select members.name,borrowed.mem_no from members right join borrowed on borrowed.mem_no=members.mem_no;

NAME	MEM_NO
Bob	2
charlie	3
	5

SQL> select members.name,borrowed.mem_no from members full join borrowed on borrowed.mem_no=members.mem_no;

NAME	MEM_NO
alice	
Bob	2
charlie	3
David	5

SQL>

VEL TECH	
EX NO.	5
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOICE (5)	5
RECORD (5)	5
TOTAL (20)	20
SIGN WITH DATE	